Lecture 0

Mathematical preliminaries

Objectives. *An observable in infinite dimension may act only on a dense subspace. This lecture defines its domain, adjoint and self-adjointness. A diagonal example distinguishes symmetry from self-adjointness. Multiplication operators prepare the construction of position and momentum in Lecture 4.*

- Determine the adjoint domain of a diagonal operator.
- Distinguish symmetric operators from self-adjoint operators.
- Explain why a real continuous multiplier is self-adjoint on its maximal domain.

0.1 Operators that cannot act on every vector

The observables of a particle need domains. Its position Q and momentum P should satisfy the *canonical commutation relation* $QP\psi - PQ\psi = \mathrm{i}\hbar\psi$ for the vectors ψ on which both sides are defined. By *Quantum Mechanics I*, Prop. 9.8, no pair of $n \times n$ matrices does, since the trace of $QP - PQ$ is 0 and the trace of $\mathrm{i}\hbar\mathbb{1}$ is $\mathrm{i}\hbar n$. In infinite dimension two cited theorems take over. They concern *bounded* operators and the self-adjoint ones among them (*Linear Algebra*, Defs 14.2 and 14.3).

By Moretti, Prop. 11.32, no two bounded self-adjoint operators Q and P on a Hilbert space satisfy $QP - PQ = \mathrm{i}\hbar\mathbb{1}$.

By the *Hellinger–Toeplitz theorem* (Teschl, Thm 2.9), an operator A that is defined on every vector and symmetric, that is, $\langle\varphi, A\psi\rangle = \langle A\varphi, \psi\rangle$ for all φ and ψ, is bounded. Observables are symmetric, as in *Quantum Mechanics I*. So Q and P cannot both be defined on every vector, and they satisfy the canonical relation only on part of the space (Lecture 4).

The energy of a particle in a box is another example. Lecture 1 shows that it multiplies a sequence of unit vectors by numbers that grow without bound. So it is not bounded, and by the Hellinger–Toeplitz theorem it cannot be a symmetric operator defined on every vector. In infinite dimension, an operator comes with the set of vectors it acts on.

0.2 Operators with domains

Definition 0.1 (Operator; densely defined)**.** An *operator* A on a Hilbert space $\mathcal{H}$ is a linear map from a subspace $D(A) \subseteq \mathcal{H}$, its *domain*, into $\mathcal{H}$. It is *densely defined* if every vector of $\mathcal{H}$ is the limit of a sequence in $D(A)$. Two operators are equal when they have the same domain and the same action on it.

Definition 0.2 (Adjoint)**.** Let A be a densely defined operator on $\mathcal{H}$. Its *adjoint* $A^\dagger$ has

the domain

$$D(A^\dagger) = \big\{\varphi \in \mathcal{H} : \text{ there is } \chi \in \mathcal{H} \text{ with } \langle \varphi, A\psi\rangle = \langle \chi, \psi\rangle \text{ for every } \psi \in D(A)\big\}, \quad (0.1)$$

and on it $A^\dagger\varphi = \chi$.

The density of the domain makes χ unique. Suppose that χ_1 and χ_2 both satisfy the condition in (0.1), and let $v = \chi_1 - \chi_2$. Then $\langle v, \psi\rangle = 0$ for every $\psi \in D(A)$. Since $D(A)$ is dense, there is a sequence ψ_j in $D(A)$ with $\psi_j \to v$. By the Cauchy–Schwarz inequality,

$$\|v\|^2 = \big|\langle v, v\rangle - \langle v, \psi_j\rangle\big| = \big|\langle v, v - \psi_j\rangle\big| \le \|v\|\,\big\|v - \psi_j\big\| \longrightarrow 0.$$

So $v = 0$, and $\chi_1 = \chi_2$. This is where the definition needs A to be densely defined.

The adjoint is an operator. Indeed, let φ_1 and φ_2 lie in $D(A^\dagger)$, with the vectors χ_1 and χ_2, and let a and b be numbers. Since the inner product is antilinear in its first slot,

$$\langle a\varphi_1 + b\varphi_2, A\psi\rangle = \overline{a}\,\langle \chi_1, \psi\rangle + \overline{b}\,\langle \chi_2, \psi\rangle = \langle a\chi_1 + b\chi_2, \psi\rangle \quad \text{for every } \psi \in D(A).$$

So $a\varphi_1 + b\varphi_2$ lies in $D(A^\dagger)$, and $A^\dagger(a\varphi_1 + b\varphi_2) = a\chi_1 + b\chi_2$.

On $\mathbb{C}^n$ the adjoint is the conjugate transpose. More generally, if A is bounded and $D(A) = \mathcal{H}$, the adjoint of *Linear Algebra*, Def. 14.3, gives a vector χ for every φ, and $A^\dagger$ is that adjoint. For other operators χ need not exist for every φ, and $D(A^\dagger)$ records the φ for which it does.

Definition 0.3 (Symmetric; self-adjoint)**.** A densely defined operator A is *symmetric* if $\langle \varphi, A\psi\rangle = \langle A\varphi, \psi\rangle$ for all $\varphi, \psi \in D(A)$. It is *self-adjoint* if $A^\dagger = A$, that is, if $D(A^\dagger) = D(A)$ and $A^\dagger\varphi = A\varphi$ there.

If A is symmetric and $\varphi \in D(A)$, then $\chi = A\varphi$ satisfies the condition in (0.1). So $D(A) \subseteq D(A^\dagger)$, with $A^\dagger\varphi = A\varphi$ there. Self-adjointness asks in addition that the adjoint have no larger domain. On $\mathbb{C}^n$ the two notions coincide, and both are the self-adjoint matrices of *Quantum Mechanics I*.

The adjoint of a symmetric operator contains the operator. Containing is a relation between operators that the course uses often.

Definition 0.4 (Extension)**.** Let A and B be operators on $\mathcal{H}$. We write $A \subseteq B$, and say that B *extends* A, when

$$D(A) \subseteq D(B) \quad \text{and} \quad B\psi = A\psi \quad \text{for every } \psi \in D(A).$$

So a symmetric A satisfies $A \subseteq A^\dagger$, and it is self-adjoint when also $A^\dagger \subseteq A$.

Proposition 0.5 (Adjoints reverse inclusions)**.** *Let A and B be densely defined operators with $A \subseteq B$. Then $B^\dagger \subseteq A^\dagger$.*

Proof. Let $\varphi \in D(B^\dagger)$, with $\chi = B^\dagger\varphi$. Then $\langle \varphi, B\psi\rangle = \langle \chi, \psi\rangle$ for every $\psi \in D(B)$. Every $\psi \in D(A)$ lies in $D(B)$, with $B\psi = A\psi$. So $\langle \varphi, A\psi\rangle = \langle \chi, \psi\rangle$ for every $\psi \in D(A)$. Hence $\varphi \in D(A^\dagger)$, and $A^\dagger\varphi = \chi = B^\dagger\varphi$. ∎

A larger operator has to satisfy the condition (0.1) for more vectors ψ, so its adjoint has a smaller domain. The following example shows a symmetric operator whose adjoint is strictly larger.

Example 0.6 (A diagonal operator on finite sequences)**.** Work in the space ℓ^2 of square-summable sequences $x = (x_1, x_2, \dots)$, with $\langle x, y\rangle = \sum_n \overline{x_n}\, y_n$ (*Linear Algebra*, Def. 13.15). Let e_n be the sequence with 1 in place n and 0 elsewhere.

1. *The operator.* Let $D(D_0)$ be the finite sequences, those with only finitely many nonzero entries, and put
$$(D_0 x)_n = n\, x_n.$$
The truncations $(x_1, \dots, x_M, 0, 0, \dots)$ of any $x \in \ell^2$ are finite sequences, and they converge to x, since $\sum_{n>M} |x_n|^2 \to 0$. So D_0 is densely defined.
2. *Symmetric.* For finite sequences x and y, the sums are finite, and $\langle x, D_0 y\rangle = \sum_n \overline{x_n}\, n y_n = \langle D_0 x, y\rangle$.
3. *The adjoint.* Let $\varphi \in D(D_0^\dagger)$, with $\chi = D_0^\dagger \varphi$. Test the condition (0.1) on $\psi = e_n$. The left side is $\langle \varphi, D_0 e_n\rangle = n\overline{\varphi_n}$, and the right side is $\langle \chi, e_n\rangle = \overline{\chi_n}$. So $\chi_n = n\varphi_n$ for every n, and since $\chi \in \ell^2$,
$$\sum_n n^2 |\varphi_n|^2 = \|\chi\|^2 < \infty.$$
Conversely, if $\sum_n n^2 |\varphi_n|^2 < \infty$, then $\chi = (n\varphi_n)$ lies in ℓ^2 and $\langle \varphi, D_0 \psi\rangle = \langle \chi, \psi\rangle$ for every finite ψ. Hence
$$D(D_0^\dagger) = \Big\{ \varphi \in \ell^2 : \sum_n n^2 |\varphi_n|^2 < \infty \Big\}, \qquad (D_0^\dagger \varphi)_n = n\, \varphi_n.$$
4. *Not self-adjoint.* The sequence $\varphi_n = 1/n^2$ has $\sum_n n^2 |\varphi_n|^2 = \sum_n 1/n^2 < \infty$, so it lies in $D(D_0^\dagger)$. It is not a finite sequence. So $D(D_0) \neq D(D_0^\dagger)$, and D_0 is symmetric and not self-adjoint.

The test on the vectors e_n in the third step fixed every coefficient of χ. Lecture 1 applies the same test to the energy of a particle in a box.

Self-check 0.2

1. Show that $\langle \psi, A\psi\rangle$ is real for every $\psi \in D(A)$ when A is symmetric.
2. Does the sequence $\varphi_n = 1/n$ lie in ℓ^2? In $D(D_0^\dagger)$?

0.3 How this course proves

Every result of this course carries one of three tags. A result with no tag is *proved* in full. A result tagged *sketched* has its central idea proved and routine details omitted, or it is proved with the help of a cited theorem. A result tagged *cited* is taken from a named source, with its page.

A few facts of analysis are used throughout. *Linear Algebra* states them without proof, and this course takes them as *standing facts*:

S1 *Plancherel*: the Fourier transform is unitary on $L^2(\mathbb{R}^n)$, with its inversion formula (Lecture 4).

S2 The Fourier basis $e^{2\pi i k x/L}/\sqrt{L}$ ($k \in \mathbb{Z}$) and the sine basis $\sqrt{2/L}\,\sin(n\pi x/L)$ ($n \geq 1$) of $L^2(0, L)$ are complete (Lecture 1 and Lecture 3).

S3 Products of orthonormal bases are orthonormal bases of $L^2(\mathbb{R}^2)$ and $L^2(\mathbb{R}^3)$ (Lecture 13).

S4 The smooth functions that vanish outside a closed bounded set inside the interval, or inside $\mathbb{R}^n$, are dense in L^2 of the interval, or of $\mathbb{R}^n$. So are the rapidly decreasing functions of Lecture 4.

A result whose only unproved ingredients are standing facts counts as proved.

Self-check 0.3

1. A proof uses the sine basis theorem and a theorem cited from Teschl. Which tag does the result carry?

0.4 Multiplication operators

The diagonal operator of Example 0.6 multiplies the nth entry of a sequence by n. Its counterpart on a line multiplies a function by a function, and the position of a particle on a line multiplies by x (Lecture 4). The space $L^2(\mathbb{R})$ is defined in the same way as L^2 of an interval (*Linear Algebra*, Def. 13.15): its vectors are classes of square-integrable functions on $\mathbb{R}$, and two functions lie in the same class when they agree outside a null set. For a bounded real multiplier on an interval, *Linear Algebra*, Ex. 14.6, shows that multiplication is a bounded self-adjoint operator. An unbounded multiplier, such as x on $\mathbb{R}$, needs a domain.

Lemma 0.7 (Multiplication operators). *Let g be a real continuous function on $\mathbb{R}$, and let M_g act by $(M_g\psi)(x) = g(x)\psi(x)$ on $D(M_g) = \{\psi : g\psi \in L^2(\mathbb{R})\}$. Then M_g is self-adjoint. If g takes a value λ at only finitely many points, then λ is not an eigenvalue of M_g.*

Proof. Write 1_n for the indicator of $[-n, n]$. On $[-n, n]$ the continuous g is bounded.

1. *Densely defined.* For $\psi \in L^2(\mathbb{R})$, the vector $1_n\psi$ lies in $D(M_g)$. By the standing fact S4, for every $\epsilon > 0$ there is a smooth f that vanishes outside some $[-n_0, n_0]$ with $\|\psi - f\| < \epsilon$. For $n \geq n_0$, $\|\psi - 1_n\psi\| = \|(1 - 1_n)(\psi - f)\| \leq \|\psi - f\| < \epsilon$. So $1_n\psi \to \psi$.
2. *Symmetric*, since g is real.
3. *The adjoint.* Let $\varphi \in D(M_g^\dagger)$ and $\chi = M_g^\dagger\varphi$. The vector $\psi = 1_n(g\varphi - \chi)$ lies in $D(M_g)$, so by Definition 0.2
$$0 = \langle \varphi, M_g\psi\rangle - \langle \chi, \psi\rangle = \int_{-n}^{n} |g\varphi - \chi|^2 \, \mathrm{d}x.$$
So $1_n g\varphi = 1_n\chi$ for every n: the functions $g\varphi$ and χ agree on every $[-n, n]$, hence on $\mathbb{R}$, and $g\varphi$ is the class χ, which lies in $L^2(\mathbb{R})$. So $\varphi \in D(M_g)$ and $M_g^\dagger\varphi = M_g\varphi$. With the symmetry, $M_g^\dagger = M_g$.
4. *No eigenvalue.* Let $g\psi = \lambda\psi$. Let J be a closed bounded interval that contains none of the points where $g = \lambda$. On J the continuous $|g - \lambda|$ has a positive minimum c, so $c^2 \int_J |\psi|^2 \leq \int_J |(g - \lambda)\psi|^2 = 0$. The open intervals between consecutive points where $g = \lambda$, and the two outside them (or $\mathbb{R}$ itself, if there are no such points), are unions of such J, so ψ vanishes on each. They cover $\mathbb{R}$ except finitely many points, which do not change a class, so $\psi = 0$. ∎

Unlike D_0, which acts only on the finite sequences, M_g acts on every ψ for which $g\psi$ is square-integrable. A unitary change of variables keeps both conclusions of Lemma 0.7.

Lemma 0.8 (Unitary equivalence). *Let U be unitary and A self-adjoint on $\mathcal{H}$. The operator $B = U^{-1}AU$, on $D(B) = \{\psi : U\psi \in D(A)\}$, is self-adjoint, and it has the same eigenvalues as A.*

Proof. For $\psi \in \mathcal{H}$, take $a_j \in D(A)$ with $a_j \to U\psi$. Then $U^{-1}a_j \in D(B)$ and $U^{-1}a_j \to \psi$, so B is densely defined. Since $\langle U\varphi, U\psi\rangle = \langle \varphi, \psi\rangle$, and $\eta = U\psi$ runs through $D(A)$ as ψ runs through $D(B)$, the condition $\langle \varphi, B\psi\rangle = \langle \chi, \psi\rangle$ for every $\psi \in D(B)$ reads $\langle U\varphi, A\eta\rangle = \langle U\chi, \eta\rangle$

for every $\eta \in D(A)$. So $\varphi \in D(B^\dagger)$ exactly when $U\varphi \in D(A^\dagger) = D(A)$, with $UB^\dagger\varphi = AU\varphi$. Hence $B^\dagger = B$. Finally, $B\psi = \lambda\psi$ if and only if $AU\psi = \lambda U\psi$. ∎

Lecture 4 applies the two lemmas with $g(x) = x$ for the position, and with the Fourier transform as U for the momentum and the energy.

Self-check 0.4

1. Let g be the constant function c. What is M_g, and which numbers are its eigenvalues? Which hypothesis of Lemma 0.7 fails?

Summary.

- *Domains are needed*: a symmetric operator defined on every vector is bounded. The position and the momentum of a particle cannot both be bounded, and the energy of a particle in a box is not (§ 0.1).
- *An operator* comes with its domain, and two operators are equal only when their domains are (Definition 0.1).
- *The adjoint* of a densely defined operator is defined on the vectors φ for which some χ satisfies $\langle \varphi, A\psi\rangle = \langle \chi, \psi\rangle$ for every $\psi \in D(A)$ (Definition 0.2).
- *A symmetric operator* A satisfies $A \subseteq A^\dagger$. It is *self-adjoint* when the two are equal (Definition 0.3), and adjoints reverse inclusions (Proposition 0.5). The diagonal operator on finite sequences is symmetric, and its adjoint is larger (Example 0.6).
- *Multiplication* by a real continuous function, on the vectors it keeps square-integrable, is self-adjoint, and a value taken at only finitely many points is not an eigenvalue. A unitary equivalence keeps both (Lemmas 0.7 and 0.8).

Lecture 1 uses these definitions for the energy of a particle in a box, and Lecture 4 uses the multiplication operators for a particle on a line.

Self-check answers.

0.2 1. Symmetry with $\varphi = \psi$ gives $\langle \psi, A\psi\rangle = \langle A\psi, \psi\rangle = \overline{\langle \psi, A\psi\rangle}$. 2. Yes, since $\sum_n 1/n^2 < \infty$. No, since $\sum_n n^2/n^2$ diverges.

0.3 1. *Sketched*: the sine basis is a standing fact, but the theorem from Teschl is not.

0.4 1. $M_g\psi = c\psi$ for every $\psi \in L^2(\mathbb{R})$, so $M_g = c\mathbb{1}$. Every nonzero vector is an eigenvector with the eigenvalue c, and by Lemma 0.7 no other number is an eigenvalue, since g takes no other value. The function takes the value c at every point, not at finitely many.

References

Moretti, *Spectral Theory and Quantum Mechanics*, Prop. 11.32 (p. 627). Teschl, *Mathematical Methods in Quantum Mechanics*, Thm 2.9 (p. 67). *Linear Algebra for Physics*, Def. 13.15, Defs 14.2 and 14.3, Ex. 14.6. *Quantum Mechanics I*, Prop. 9.8.